Loss of TP53 accelerates growth of intestinal organoids through mitochondrial remodeling

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Abstract

Intestinal organoids recapitulate many features of the adult epithelium, yet the metabolic consequences of tumor-suppressor loss remain poorly understood. Here we used CRISPR-Cas9 to delete TP53 in mouse intestinal organoids and combined live imaging with respirometry. TP53-deficient organoids grew 35% faster than controls ($p < 0.01$) and displayed elongated mitochondria with increased oxygen consumption. These results suggest that mitochondrial remodeling contributes to the early proliferative advantage of TP53 loss.

1 Introduction

Organoids derived from adult stem cells provide a tractable model of epithelial homeostasis and disease. Mutations in TP53 are among the most frequent alterations in colorectal cancer, but their immediate effects on cellular metabolism are difficult to isolate in vivo [1, 2].

Mitochondria integrate nutrient availability with growth signaling. Previous work has shown that p53 regulates the expression of genes involved in oxidative phosphorylation, suggesting a direct link between tumor suppression and energy metabolism [3].

2 Materials and Methods

Organoids were derived from the small intestine of C57BL/6 mice and maintained in Matrigel with standard growth factors. Guide RNAs targeting exon 5 of Trp53 were delivered by electroporation, and clones were validated by sequencing.

Oxygen consumption was measured with a Seahorse analyzer. Growth was quantified as the change in organoid area over 72 hours, and statistical significance was assessed with a two-sided t-test.

$$G = (A_{72} - A_0) / A_0$$

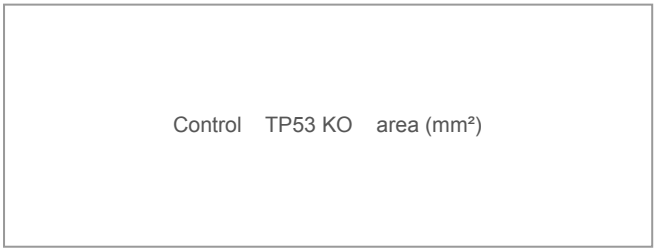


Figure 1: Growth of control and TP53-deficient organoids over 72 hours. Bars show mean ± SD of three independent experiments.

3 Results

TP53-deficient organoids expanded significantly faster than controls and formed more crypt-like buds. Electron microscopy revealed elongated mitochondria with dense cristae, and basal respiration increased by 28% relative to controls.

Inhibition of mitochondrial fission partially reversed the growth advantage, indicating that mitochondrial dynamics are required for the phenotype.

4 Discussion

Our findings identify mitochondrial remodeling as an early consequence of TP53 loss in intestinal epithelium. Whether similar changes occur in human organoids and in tumors in vivo remains to be determined.

References

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3. Sample D. p53 and oxidative phosphorylation. Mito Lett. 2018;7:5–9.